

ELK421E GENERATION OF ELECTRICAL ENERGY

Name & Family Name:	Q1	Q2	Q3	Q4	Prof. Dr. Ömer Usta Kemal Aygöl
Number:					
Signature:					
Date: 06.02.2021					
Instructions +This Open Book examination paper consists of 4 questions. +You are not permitted to discuss or take assistance of others to solve the examination problems. +You would be severely penalized if your solutions appear to have been copied from others. +Your solutions should be methodical. You would be penalized if your solutions are illegible. +Please use your own statements while answering the questions					

Final (Open Book) Exam

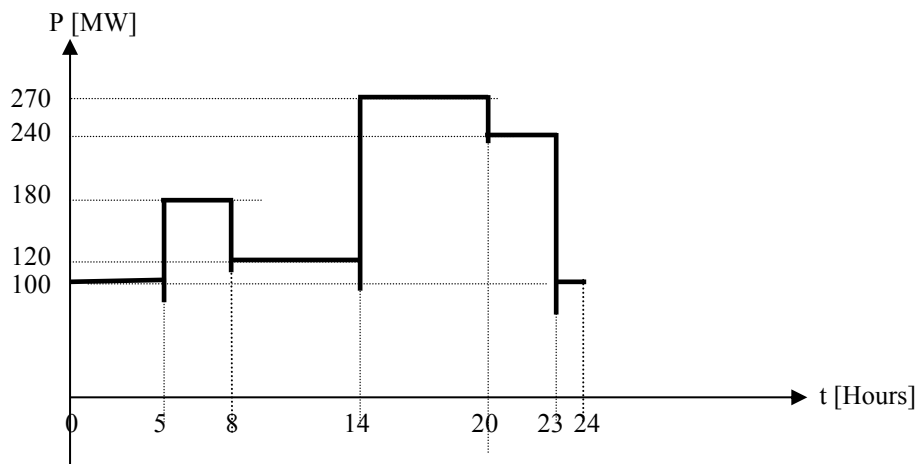
PROBLEM 1[25pts]: A Hydro power plant has the following daily discharges for a week and also following data. The plant has an effective head of 20 m and an overall efficiency of 87%. If the plant operates on 42% load factor, estimate (i) the average daily discharge (ii) pondage required and (iii) installed capacity of proposed plant.

Days	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
Q(m ³ /s)	600	750	800	810	850	650	575

PROBLEM 2[25pts]: A steam plant is operating in parallel with a hydro (pomp-reservoir) plant. The daily load curve is seen following figure, and is the same for each day of the year. The head of the hydro plant is 116m and the net discharge of the plant 35 m³/s

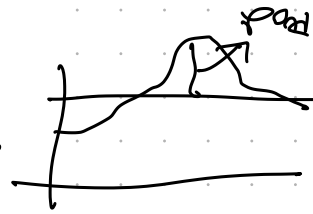
- Find out the yearly duration curve.
- Find out the average price of generated unit.

	Steam	Hydro-Pomp
Capital cost [TL/kW]	2000	4000
Interest [%]	12	12
Fuel price [TL/ton]	185	-
Calorific value [kcal/kg]	4000	-
Efficiency [%]	36	-
Electric efficiency [%]	-	94
Turbine efficiency [%]	-	80
Hydro-Pomp efficiency [%]	-	75



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Q'ları topla 7'ye böl → ① $Q \rightarrow m^3/s$

$$\eta \cdot m \cdot g \cdot h = J/s = W$$

↳ 1 ton su
1000 kg/s

$$LF = \frac{\text{Aer load (W)}}{\text{Max D. (W)}} = 0.42 \text{ ? } \rightarrow \text{buru gettin}$$

Pondage için $\text{avege debi} = Q'ları \text{ topla } 7'ye \text{ böl}$

850 - avege debi = debi pondage
↳ kW gelir

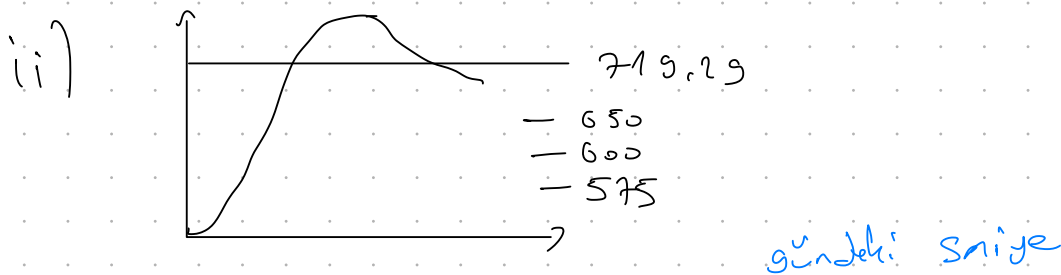
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$$LF = \frac{\text{avg load}}{\text{max demand}}$$

Days	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
Q(m ³ /s)	600	750	800	810	850	650	575

$$h = 20 \text{ m} \quad \eta = \% 87 \quad LF = \% 42$$

$$\begin{aligned} \text{i) avg daily discharge} &= \frac{600 + 750 + 800 + 810 + 850 + 650 + 575}{7} \\ &= 719,29 \text{ m}^3/\text{s} \end{aligned}$$



$$\frac{(650 + 600 + 575) \text{ m}^3}{3} \cdot 24 \cdot 60 \cdot 60 = 15768 \cdot 10^4 \text{ m}^3$$

$$3 \cdot 719,29 \cdot 24 \cdot 60 \cdot 60 = 186439968 \text{ m}^3$$

$$\text{pondage required} = 186439968 - 15768 \cdot 10^4$$

$$= 28759968 \text{ m}^3$$

$$\text{iii) } 719,29 \text{ m}^3/\text{s} = 719,29 \cdot 10^3 \text{ kg/s}$$

$$1 \text{ W.s} = 1 \text{ J} \quad \text{J/s} = \text{W}$$

$$\text{avg power produced} = 719,29 \cdot 10^3 \frac{\text{kg}}{\text{s}} \cdot 9,81 \cdot 20 \cdot 0,87$$

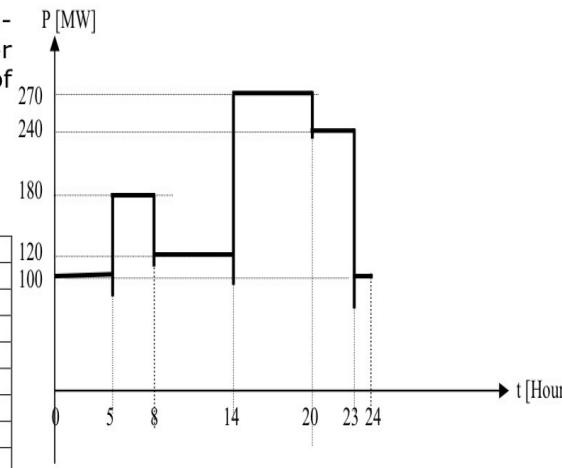
$$= 122778487,3 \text{ W}$$

$$LF = \frac{\text{avg load}}{\text{max demand}} \quad 0,42 = \frac{122778487,3 \text{ W}}{\text{max demand}}$$

$$\text{max demand} = 292329731,6 \text{ W}$$

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- (a) Find out the yearly duration curve.
 (b) Find out the average price of generated unit. TL?



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Total
 Average Power [MW] = $\frac{5 \cdot 100 + 3 \cdot 180 + 6 \cdot 120 + 6 \cdot 270 + 3 \cdot 240 + 1 \cdot 100}{24}$
 = 175 MW

$35 \text{ m}^3/\text{s} = 35 \cdot 10^3 \text{ kg/s}$ $1 \text{ m}^3 = 1 \text{ ton}$ su $1 \text{ W.s} = 1 \text{ J}$
 $\text{W} = \text{J/s}$

hydro power = $\frac{35000 \text{ kg/s}}{\text{M}} \cdot \frac{9,82}{\text{g}} \cdot \frac{1}{\text{h}} \cdot \frac{\text{efficiency}}{\text{efficiency}} = 22486228,8 \text{ W}$
 = 22,486 MW

hydro power = 22,486 MW steam power = $175 - 22,486$
 = 152,514 MW

steam interest = $152,514 \cdot 10^3 \text{ kW} \cdot 2000 \frac{\text{TL}}{\text{kW}} \cdot \frac{12}{100} = 36603360 \$$

hydro interest = $22,486 \cdot 10^3 \text{ kW} \cdot 4000 \cdot \frac{\text{TL}}{\text{kW}} \cdot \frac{12}{100} = 10793280 \$$

$(1 \text{ kWh} = 860 \text{ kcal})$ $1 \text{ MWh} = 860 \cdot 10^3 \text{ kcal}$

steam power energy = $152,514 \cdot 8760 = 1336022,64 \text{ MWh} = 1,14 \cdot 10^{12} \text{ kcal}$

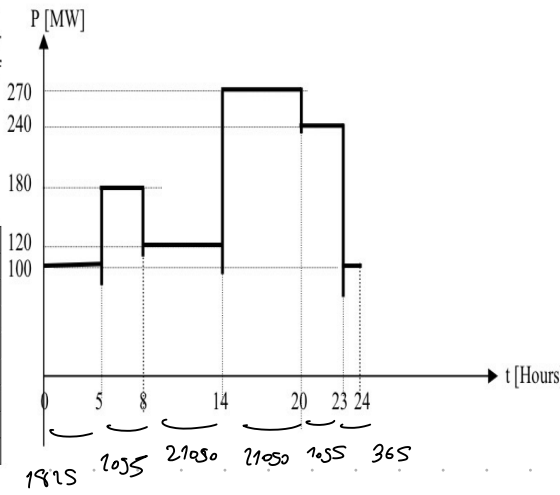
$\frac{1,14 \cdot 10^{12} \text{ kcal} \cdot \frac{100}{36}}{4000 \text{ kcal/kg}} = 79166666,7 \text{ kg}$

$\frac{185 \text{ TL}}{\text{ton}} \cdot 79166666,7 \text{ ton} =$
 = 146458333,3 TL

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$$\frac{5 \text{ saat}}{365 \text{ gün}} \cdot 365 \text{ gün} = 1825 \text{ saat}$$

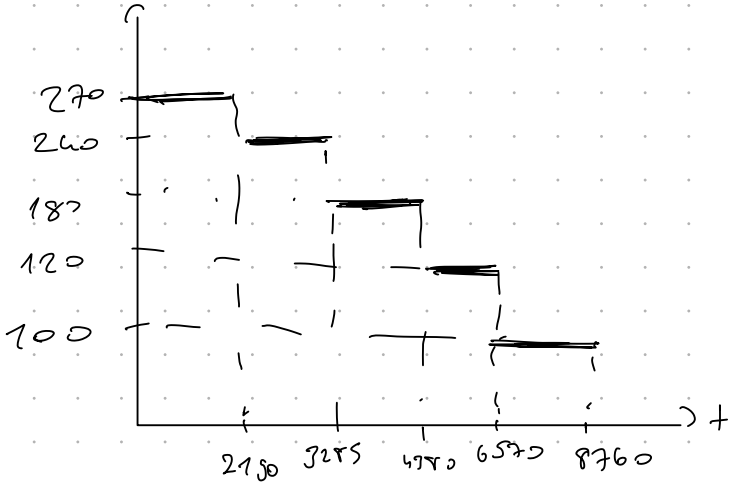
$$3 \cdot 365 = 1095 \text{ saat}$$

$$6 \cdot 365 = 2190 \text{ saat}$$

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$$3 \cdot 365 = 1095 \text{ ''}$$

$$1 \cdot 365 = 365 \text{ ''}$$



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PROBLEM 3[25pts]: A factory load consists of the following:

- (i) an induction motor of 150 h.p. (111.9 kW) with 0.7pf lagging and 80% efficiency;
- (ii) a synchronous motor of 100 hp(74.6 kW) with 0.85 pf leading at 90% efficiency
- (iii) a lighting load of 50 kW.

Find the annual electric charges, if the tariff is 100\$ per annum per kVA maximum demand plus 7 cent per kWh; assuming the load to be steady throughout the year.

PROBLEM 4[30pts]:

(a). What do you mean by system grounding or earthing? What are the methods of grounding and explain their effects on the system protection.

(b). A 3-phase, 20 MVA, 11kV star connected alternator is protected by Percentage Current Differential Relay. The star point is earthed through a resistance of 5 ohms. If the CTs have a ratio of 1000/5 and the relay is set to operate when there is an out of balance current of 15 A, calculate:

(i) the percentage of each phase of the stator winding which is unprotected.

(ii) the minimum value of grounding resistance to protect 90% of the winding.

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- (i) an induction motor of 150 h.p. (111.9 kW) with 0.7 pf lagging and 80% efficiency;
- (ii) a synchronous motor of 100 hp (74.6 kW) with 0.85 pf leading at 90% efficiency
- (iii) a lighting load of 50 kW. $\text{pf} = 1$

Find the annual electric charges, if the tariff is 100\$ per annum per kVA maximum demand plus 7 cent per kWh; assuming the load to be steady throughout the year.

$$\text{tariff} = 100 \$ / \text{kVA} + 0,07 \$ / \text{kWh}$$

$$\text{Total } S(\text{kVA}), P(\text{kW})$$

$$S = \sqrt{P^2 + Q^2}$$

$P = \text{Real Power (kW)}$
 $Q = \text{Reactive Power (kVAr)}$
 $S = \text{Complex Power (kVA)}$

$$\textcircled{1} \quad P = S \cdot 0,7 \quad P = 111,9 \cdot \frac{100}{80} = 139,875 \text{ kW}$$

$$139,875 = S \cdot 0,7 \quad S = 199,82 \text{ kVA}$$

$$\textcircled{2} \quad P = S \cdot 0,85 \quad P = 74,6 \cdot \frac{100}{90} = 82,88 \text{ kW}$$

$$82,88 = S \cdot 0,85 \quad S = 97,5 \text{ kVA}$$

$$\textcircled{3} \quad P = 50 \text{ kW} \quad S = 50 \text{ kVA}$$

$$\text{Total } S(\text{kVA}) = 199,82 + 97,5 + 50 = 347,32 \text{ kVA}$$

$$\text{Total } P(\text{kW}) = 139,875 + 82,88 + 50 = 272,75 \text{ kW}$$

$$\text{tariff} = 100 \$ / \text{kVA} + 0,07 \$ / \text{kWh}$$

$$\text{Total (kWh)} = 272,75 \cdot 8760 = 2389290 \text{ kWh}$$

$$\text{Total } \$ = 100 \$ / \text{kVA} \cdot 347,32 \text{ kVA} + 0,07 \$ / \text{kWh} \cdot 2389290 \text{ kWh}$$

$$\text{Total } \$ = 201982,3 \$$$

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(b). A 3-phase, 20 MVA, 11kV star connected alternator is protected by Percentage Current Differential Relay. The star point is earthed through a resistance of 5 ohms. If the CTs have a ratio of 1000/5 and the relay is set to operate when there is an out of balance current of 15 A, calculate:

- (i) the percentage of each phase of the stator winding which is unprotected. $\%X$?
 (ii) the minimum value of grounding resistance to protect 90% of the winding. r ?

$$r = 5 \Omega$$

$$V_{ph} = \frac{11}{\sqrt{3}} = 6,35 kV$$

$$I_{min} = 15A$$

$$V_x = V_{ph} \cdot \frac{X}{100} = 6,35 \cdot 10^3 \cdot \frac{X}{100} = 63,5X$$

$$\frac{63,5 \cdot X}{r} = I_{min}$$

$$\frac{63,5 \cdot X}{5} = 15$$

$$X = 1,18 \rightarrow \%1,18$$

$$\%X = \%10$$

$$15 = \frac{63,5 \cdot 10}{r}$$

$$r = 42,33 \Omega$$